

controller-based control system with firmware to control the motor is embedded within the BLDC motor for better control and protection. This electronic controller detects the rotor position and ensures a reliable start to ceiling fan, eliminating the requirement of sensors and resulting in a cost-effective design. The controller detects the speed commanded by the user by processing the magnitude and timing of the changes in internal DC bus voltage that occurs when the wall-mount power switch is switched on/off. These changes are interpreted by the microcontroller appropriately and the fan is run at the corresponding speed. The motor also embeds an infrared receiver that enables remote control of the ceiling fan. The fan is supplied power in the same way as a standard ceiling fan and hence there is not need to modify the existing wiring method.

It took three years for Sundar and his

team to come up with the Superfan and three different patents are filed for these efficient-design technologies used in the commercially-available product.

It is also interesting to note that Superfan can be used with other appliances to add to their efficiency. Sunder says, "The savings from Superfan are 50 per cent at top speed, but as most of the fans run at medium speed, it converts into more savings. Superfan has excellent-quality bearings, which enhance its life and can be used with air-conditioners to save a lot of money." If you normally set your AC at 21°C, with the fan running you may set it at 23°C and yet feel the same kind of cooling effect. For every one degree of higher setting of the AC, roughly three per cent power is saved.

Superfan, by design, works in a wide voltage range and comes in three variants, in different colours and curves, with and without sup-

port of an infrared remote controller. These variants are basically designed for different applications in different room types. The quiet SuperA1 model is designed for hospitals, classrooms, workspaces, AC rooms, bedrooms whilst SuperX1/X7 models are suitable for high-speed, high-air delivery in individual rooms and workspaces. Also, there are SuperV1/V7 models for wide air delivery in banquet halls, conference rooms, etc.

Future

Superfan is the first of its kind in the market today. Speaking about future plans, Sunder says, "Versa Drives is keen on developing more efficient ceiling and other types of fans and other appliances with energy efficiency and intelligence."

Superfan can be bought from superfan.in ●

The author is a technical journalist at EPY, Gurgaon

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